Listing 4.1 Creating and terminating a virtual machine from the CLI (Bash)

```
#!/bin/bash -e                                                    ①
AMIID="$(aws ec2 describe-images --filters \                      ②
➥ "Name=name,Values=amzn2-ami-hvm-2.0.202*-x86_64-gp2" \
➥ --query "Images[0].ImageId" --output text)"
VPCID="$(aws ec2 describe-vpcs --filter "Name=isDefault,
➥ Values=true" \                                                 ③
➥ --query "Vpcs[0].VpcId" --output text)"
SUBNETID="$(aws ec2 describe-subnets --filters \                  ④
➥ "Name=vpc-id, Values=$VPCID" --query "Subnets[0].SubnetId" \
➥ --output text)"
INSTANCEID="$(aws ec2 run-instances --image-id "$AMIID" \         ⑤
➥ --instance-type t2.micro --subnet-id "$SUBNETID" \
➥ --iam-instance-profile "Name=ec2-ssm-core" \
➥ --query "Instances[0].InstanceId" --output text)"
echo "waiting for $INSTANCEID ..."
aws ec2 wait instance-running --instance-ids "$INSTANCEID"        ⑥
echo "$INSTANCEID is up and running"
echo "connect to the instance using Session Manager"
echo "https:/ /console.aws.amazon.com/systems-manager/
➥ session-manager/$INSTANCEID"
read -r -p "Press [Enter] key to terminate $INSTANCEID ..."
aws ec2 terminate-instances --instance-ids
➥ "$INSTANCEID" > /dev/null                                      ⑦
echo "terminating $INSTANCEID ..."
aws ec2 wait instance-terminated --instance-ids
➥ "$INSTANCEID"                                                  ⑧
echo "done."
```

① -e makes Bash abort if a command fails.

② Gets the ID of Amazon Linux AMI

③ Gets the default VPC ID

④ Gets the default subnet ID

⑤ Creates and start the virtual machine

⑥ Waits until the virtual machine is started

⑦ Terminates the virtual machine

⑧ Waits until the virtual machine is terminated



Cleaning up